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Greedy Exploration Failure, Sampling, and Regret Bounds in Bandits

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Introduction

In bandits, a greedy method can lock onto the wrong arm if early rewards are misleading. The main issue is not just the long-run limit, but how much sampling happens early and what regret the method pays before it settles. (LLM Memory)

The core tension in multi-armed bandits is simple: an algorithm must use the arm that looks best now, but it also must keep testing arms that may be better than they first appear (Model-Generated). A purely greedy rule often fails because early random outcomes can make a bad arm look good, after which the algorithm keeps exploiting it and stops gathering the evidence needed to correct the mistake (Model-Generated). This creates suboptimal convergence: the method can converge to always picking a suboptimal arm with nonzero probability, especially when some arms are sampled too little at the start (Model-Generated). This is why bandit analysis focuses on both regret bounds and sampling behavior. Regret captures the cost of wrong choices over time, while finite-time analysis shows whether an algorithm is reliable before asymptotic guarantees become meaningful (Model-Generated). The key question behind the query is therefore how exploration failure, insufficient sampling, and greedy lock-in connect to formal regret guarantees and to the practical trade-off between fast exploitation and safe learning (Model-Generated).

Exploration Failure and Greedy Suboptimal Convergence

Greedy fails because one bad early draw from the best arm can stop the algorithm from ever returning to it. This is not a rare edge case: many papers show a positive-probability failure event that leads to linear regret. (11 sources)

A clean way to see exploration failure is that greedy uses current estimates as if they were already reliable. If early noise makes a bad arm look best, the algorithm keeps pulling that arm and does not collect the counterevidence needed to fix the error. In simple Bernoulli bandits, this can happen immediately: if the good arm first gives 0 and another arm first gives 1, greedy may discard the best arm forever and lock into a bad one, causing regret that grows linearly in time (Jedor et al., 2021) (Slivkins et al., 2025) (Banihashem et al., 2026). This basic failure mode is widely noted: after a small warm start, follow-the-leader or greedy can leave the optimal arm underestimated and never sampled again, so convergence to the wrong arm happens with positive probability rather than only in a pathological worst case (Ortner, 2022) (Boursier et al., 2022) (Bastani et al., 2017). Recent work sharpens this point by showing that such “suffix failures” are not just due to one-sample initialization; they can persist even with many initial samples, with failure probability that can remain polynomially or even constantly large depending on the instance (Krishnamurthy et al., 2024) (Banihashem et

al., 2023) (Banihashem et al., 2026). The same negative message appears in Bayesian and contextual views: with positive constant probability, greedy may fail to try the best arm enough—or even once—and then incur a constant regret each round, that is, linear total regret (Raghavan et al., 2020) (Aridor et al., 2019). In contextual settings this risk can depend on structure: if conditions like covariate diversity do not hold, greedy can again fail to learn the true best policy and suffer linear regret some fraction of the time (Bastani et al., 2017). A useful modern interpretation is that greedy succeeds only when the problem has enough built-in identifiability to keep revealing the best arm without forced exploration; otherwise decoy arms can trap it forever (Slivkins et al., 2025). This is why the literature treats pure greedy as fundamentally unsafe, even though it may look very good on runs where the early samples happen to be favorable (Jedor et al., 2021) (Brunner et al., 2022).

Evidence

(Jedor et al., 2021)

Be Greedy in Multi-Armed Bandits

"The exploration, although detrimental in the short term, is usually needed in the worst-case as it ensures that the learning algorithm "converges" to the optimal arm in the long run. On the other hand, the GREEDY algorithm, an exploration-free strategy, focuses on pure exploitation and pulls the apparently best arm according to the information gathered thus far, at the risk of only sampling once the true optimal arm. This typically happens with Bernoulli rewards where only arms whose first reward is a 1 will be pulled again (and the others discarded forever). As a consequence, with some non-zero probability, the regret grows linearly with time"

"Example 1 Consider a relatively simple Bernoulli bandit problem consisting of K = 2 arms with expected rewards 0.9 and 0.1 respectively. With probability at least 0.01, GREEDY fails to find the optimal arm. On the other hand, with probability 0.9 2 it suffers no regret after the initial pulls. This results in a linear regret with a large variance."

(Slivkins et al., 2025)

Greedy Algorithm for Structured Bandits: A Sharp Characterization of Asymptotic Success / Failure

"A classic example is as follows: "Suppose the reward of each arm follows an independent Bernoulli distribution with a fixed mean, and Greedy is initialized with a single sample per arm. If the best arm initially returns a 0 while another arm returns a 1, Greedy permanently excludes the best arm.""

"We provide a complete characterization of when Greedy asymptotically fails (incurs linear regret) vs when it succeeds (achieves sublinear regret)"

".The negative results are of primary interest here, as they substantiate the common belief that Greedy performs poorly"

"Our negative results establish a positive (but possibly very small) constant probability of a "failure event" where Greedy gets permanently stuck on a decoy, for an infinite time horizon"

*".(b) Suppose the warm-up data consists of one sample for each arm. Assume f * has a decoy f dec ∈ F, with decoy arm a dec . Then with some probability p dec > 0 it holds that Greedy chooses a dec for all rounds t ∈ (T 0 , ∞). We can lower-bound p dec by e −O(K/Γ 2 (fdec))"*

".Thus, Greedy succeeds, in the sense of achieving sublinear regret for any warm-up data, if and only if the problem instance is self-identifiable. Else, Greedy fails for some warm-up data, incurring linear expected regret."

(Banihashem et al., 2026)

Bandit Social Learning with Exploration Episodes

"While agents are incentivized to explore within their respective episodes, we show that the aggregate exploration fails: e.g., its Bayesian regret grows linearly over time. In fact, such failure is a (very) typical case, not just a worst-case scenario"

".The social learning dynamics described above can be interpreted as a "greedy" bandit algorithm (Greedy) which only exploits. As such, it has been believed to perform poorly, so that exploration is needed"

".If the best arm initially returns a 0 while another arm returns a 1, Greedy gets permanently stuck on the bad arm. Thus, we have a type of a learning failure discussed above. Banihashem et al. (2023) extend this example in various directions, proving that similar learning failures (i) hold with substantial probability even if the greedy algorithm is initialized with many samples;"

"One take-away message is that positive-probability learning failure is the typical case for K-armed bandits."

(Ortner, 2022)

Adaptive Algorithms for Meta-Induction

"As already mentioned the greedy FTL algorithm, that after initially sampling each arm once chooses at each further step t the arm that maximizes the observed average reward r so far, is a risky choice. Indeed, the optimal arm may underperform in the first few samples taken from it, and may then not be chosen again, leading to regret linear in T. However, slight modifications work well in practice and in theory. For example, adding a bit of randomization by choosing a random arm with some small probability and following (11) with probability 1 − is sufficient to obtain a successful algorithm. While for fixed the regret remains linear, if is chosen to decrease over time one can derive strong performance bounds (Auer et al., 2002). Note that the added randomization is dealing with the exploration-exploitation dilemma in a very direct way with being the fraction of exploration the algorithm conducts."

(Boursier et al., 2022)

A survey on multi-player bandits

"When ε t = 0 for all t, it is called the greedy algorithm, as it always greedily pulls the best empirical arm. The greedy algorithm generally incurs a linear regret in T , as the best arm can be underestimated after its first pull and never be pulled again. Appropriately choosing the sequence (ε t) instead leads to a sublinear regret, as given by Theorem 9"

".Distinctly separating the exploration and the exploitation phase leads to a larger regret bound. Especially, if all the arms are explored the same amount of time (uniform exploration), the instance dependent bound scales with 1 Δ 2 ."

(Bastani et al., 2017)

Mostly Exploration-Free Algorithms for Contextual Bandits

"In general, algorithms that fail to explore sufficiently may fail to learn the true arm parameters, yielding poor performance"

". if covariate diversity is not satisfied, the greedy algorithm may be undesirable since it may achieve linear regret some fraction of the time (i.e., it fails to converge to the optimal policy with positive probability)."

(Krishnamurthy et al., 2024)

Can large language models explore in-context?

"The Greedy algorithm (Footnote 6) is fundamentally inefficient in that it does not adaptively steer its exploration toward better-performing arms. Accordingly, its regret rate scales as T 2/3 (for an optimal setting of T −1/3). Fixing such regret does not improve for easier instances"

".The Greedy algorithm (Footnote 5) does not explore at all, which causes suffix failures. This is obvious when the algorithm is initialized with a single sample (n = 1) of each arm: a suffix failure happens when the good arm returns 0, and one of the other arms returns 1. However, suffix failures are not an artifact of small n: they can happen for any n, with probability that scales as Ω(1/ ✓ n) (Banihashem et al., 2023)."

(Banihashem et al., 2023)

Bandit Social Learning: Exploration under Myopic Behavior

"more moderate versions lead to stark exploration failures, and consequently to regret rates that are linear in the number of agents. We provide matching upper bounds on regret"

"As a special case of independent interest, we obtain a general result on failure of the greedy algorithm in multi-armed bandits."

(Raghavan et al., 2020)

Greedy Algorithm almost Dominates in Smoothed Contextual Bandits

"One way to formalize a negative result is to consider a Bayesian prior over 2-armed bandit instances. Then, with positiveconstant probability over the prior, the greedy algorithm fails to explore the best arm, and therefore incurs a positive-constant regret in each round (see Chapter 11 in [30])"

".We substantially improve over the Õ(✓ T) regret bound from Kannan et al. [19]: our main result applies per-instance rather than only in the worst-case, and allows perturbations to be correlated across actions."

(Aridor et al., 2019)

Competing Bandits: The Perils of Exploration Under Competition

"Finally, the greedy algorithm is terrible on a wide variety of problem instances, in the sense that with constant probability it fails to try the best arm even once, and therefore suffers regret Ω(T) (see Chapter 11.2 in Slivkins, 2019)."

(Brunner et al., 2022)

Reviewing Recommender Systems in the Medical Domain

> *"A greedy strategy refers to exploitation without exploring, i.e., choosing the arm with the highest current reward estimate. This bears the risk of missing the optimal arm forever"*
>
> *"However, the risk remains that the exploration continues after having identified the optimal arm, since the algorithm forces the agent to select a (known) suboptimal arm during each exploration step, thus the regret will never converge to zero."*

Insufficient Sampling and Failure Mechanisms

The main failure mechanism is simple: the algorithm does not gather enough evidence from the arms it is currently undervaluing. If exploration is too weak, beliefs formed from early noise persist; if exploration is too strong, the algorithm wastes rounds and also loses badly. (5 sources)

What turns greedy failure into a persistent problem is not just a wrong early estimate, but a lack of later samples that could overturn it. Under scarce information, a greedy learner tends to keep sampling what already looks good, which can make it miss “hidden treasure” arms that need more tries before their value becomes visible; in that sense, guided sampling can be worse than even blind uniform sampling when the good option is initially hard to detect (Peshkin et al., 2002). This is the same balancing problem emphasized in modern bandit work: cutting exploration too quickly creates false beliefs about the best arm and repeated pulls of suboptimal arms, while cutting it too slowly wastes many rounds on known-bad arms, so both sides can raise regret (Markovic et al., 2021). The mechanism is therefore an evidence bottleneck: once an arm falls behind in empirical mean, a myopic rule may stop feeding it data, and then the estimate cannot recover fast enough to change action choice.

This also explains why simple fixes like fixed-\epsilon exploration are only partial solutions. A constant exploration rate does prevent complete lock-in, but in open-ended horizons it keeps paying a permanent exploration tax, which leads to poor regret performance; at the same time, schedule-based decay is hard to tune, because exploration that fades too fast recreates the original insufficient-sampling problem (Markovic et al., 2021). More recent analyses make this decomposition explicit for \epsilon-greedy: regret comes from two channels, random exploration pulls and greedy pulls made under wrong empirical rankings. If the decay schedule supplies enough samples to each arm, the greedy-error part can be bounded tightly, but if it does not, the algorithm remains vulnerable to long periods of wrong exploitation (Li et al., 2025). In that sense, “insufficient sampling” is not a vague intuition; it is the concrete reason the greedy component stays wrong.

The same pattern appears even more clearly when the environment changes or when early beliefs become stale. If \epsilon is very small, the algorithm may need a long time just to revisit an arm that has become optimal, and if its old favorite still has a high empirical average, belief inertia can keep the algorithm exploiting the wrong arm for a linear number of rounds before enough fresh samples arrive to correct it (Mendelson et al., 2025). This shows that failure is often about sample timing as much as sample count: the needed observations arrive too late to prevent large regret. More broadly, work on myopic exploration in reinforcement learning makes the same point at a larger scale: methods like \epsilon-greedy, softmax, or simple action noise can work well in some tasks, but they can also explore very inefficiently, and their performance depends on how often the induced greedy policy remains suboptimal before enough data are collected to fix it (Dann et al., 2022). So the practical lesson is that suboptimal convergence and weak finite-time performance usually come from one root cause: the algorithm does not allocate enough informative samples, early enough, to the actions it is in danger of ruling out.

Evidence

(Peshkin et al., 2002)

Learning from Scarce Experience

> *"We would expect a greedy learning algorithm to sample near policies that look better under scarce information, tending to choose the sub-optimal a 2 in the HT problem. This strategy is inferior to blind sampling, which samples uniformly from the policy space and*

will discover the hidden treasure of a 1 faster"

". However, although acting greedily is somewhat better in HF, it is much worse in HT. This illustrates why, without any prior knowledge of the domain and given a limited number of samples, it is important not to guide sampling too much by optimization."

(Markovic et al., 2021)

An empirical evaluation of active inference in multi-armed bandits

"Naturally, this is a fine balancing act: reducing exploration too quickly would potentially result in false beliefs about the best arm, hence repeatedly choosing sub-optimal arms and accumulating regret. In contrast, reducing exploration too slowly would result in wasting too many rounds exploring sub-optimal arms and again accumulating regret"

".Note that -greedy or Softmax action-selection strategies [23], frequently used in reinforcement learning, have fixed exploration, and consequently poor regret performance in open ended bandit problems (i.e. problems with an unknown time horizon). There are variants of these strategies where exploration parameters, in -greedy and τ in Softmax, are reduced with specific schedules (Auer et al., 2002). However, choosing a schedule is based on heuristics and parameters are difficult to tune."

(Li et al., 2025)

Choosing the Better Bandit Algorithm under Data Sharing: When Do A/B Experiments Work?

"When an algorithm runs individually (left panel), too much exploration (e.g., sampling arms uniformly at random) or too much exploitation (e.g., the greedy algorithm) is highly suboptimal and leads to a linear regret"

".In contrast, an overly-exploiting algorithm leads to sign violation, because over-exploitation (and hence insufficient exploration) is only beneficial under data sharing due to free riding, but detrimental without data sharing"

".For an greedy algorithm, a suboptimal arm can be pulled in two cases: (1) an exploration step selects this arm when sampling uniformly at random; or (2) a greedy step selects this arm because its empirical mean is the highest among all arms. For the exploration steps, the number of pulls on any suboptimal arm is bounded by summing the exploration probabilities where t = Θ 1 t 1−α , incurring a regret on the order of T t=1 1 t 1−α = T α . For the greedy steps, we show that the exploration steps allow the algorithm to collect sufficient samples for each suboptimal arm, and the number of pulls from the greedy steps is thus bounded by a constant."

(Mendelson et al., 2025)

Fooling Algorithms in Non-Stationary Bandits using Belief Inertia

"The first instance addresses the case of very small during the initial rounds the algorithm identifies a single arm as optimal, and then the rewards switch drastically. Since exploration occurs at rate only K, it takes a long time before the algorithm even samples the new optimal arm, incurring large regret in the meantime. The second instance handles the remaining values of here the identity of the optimal arm changes midway through the horizon, and because of belief inertia, the empirical average of the previously optimal arm remains anchored above that of the new optimal arm. As a result, the algorithm continues to exploit the wrong arm for a linear number of rounds before sufficient exploration corrects its belief."

(Dann et al., 2022)

Guarantees for Epsilon-Greedy Reinforcement Learning with Function Approximation

"Myopic exploration policies such as epsilon-greedy, softmax, or Gaussian noise fail to explore efficiently in some reinforcement learning tasks and yet, they perform well in many others"

". This paper presents a theoretical analysis of such policies and provides the first regret and sample-complexity bounds for reinforcement learning with myopic exploration"

". The first term denotes the regret of the corresponding greedy policies, which can be controlled by invoking Theorem 1 to bound the number of episodes for which the greedy policies are suboptimal."

Regret Bounds and Asymptotic Guarantees

Regret theory makes the same point as the failure examples: if exploration is missing or badly tuned, greedy-type methods usually pay linear regret. To get sublinear regret, the algorithm must force enough sampling, but the exact rate and the finite-time cost depend a lot on how that exploration is scheduled. (7 sources)

From the regret point of view, pure greedy is bad not just because it can converge to the wrong arm, but because that lock-in directly turns into linear cumulative regret. This is now a standard benchmark in the literature: insufficient exploration means the learner does not collect enough information, so its per-round

mistake rate does not vanish fast enough to make total regret sublinear (Lee et al., 2025). The same logic explains why a fixed exploration rate is also unsatisfactory in long horizons. If \epsilon stays constant, the algorithm keeps paying a constant exploration cost forever, so regret grows linearly rather than logarithmically; but if \epsilon is set too small, the method reverts toward over-exploitation before it has learned enough (Ameur et al., 2020) (Adjif et al., 2019). This is why decaying-exploration rules matter asymptotically: with a schedule that shrinks over time, the algorithm can spend more rounds learning early and then mostly exploit later (Ameur et al., 2020) (Adjif et al., 2019).

The next step in the theory is that not all forced-exploration schedules give the same asymptotic guarantee. Some greedy-plus-exploration methods achieve only weaker sublinear rates, such as O(T^{2/3}) when the exploration probability is set around t^{-1/3}, which is enough to avoid linear regret but is still worse than logarithmic or \sqrt{T}-type performance from stronger adaptive methods (Qi et al., 2023). Recent summaries therefore describe \epsilon-greedy and simple explore-then-commit schemes as theoretically suboptimal in regret compared with more adaptive rules such as UCB- or Thompson-style methods, even though they are easy to use (Lee et al., 2025). At the same time, asymptotic order alone does not tell the whole story: too few exploration rounds can make constants blow up sharply, which means a method may be “sublinear” on paper but still perform poorly for realistic sample sizes (Lee et al., 2025).

The more refined message from recent work is that regret bounds depend not only on how much explicit exploration is added, but also on whether the problem itself creates extra sampling opportunities. In blocked or constrained settings, Basu et al. show that a greedy-UCB rule can benefit from “free exploration”: because the current best arm is sometimes unavailable, the algorithm is forced to sample other arms, and this can improve the regret bound relative to analyses that ignore that effect (Basu et al., 2019). This is useful conceptually because it separates two cases: when the environment naturally keeps information flowing, greedy components can be much safer; when it does not, the algorithm must create that information itself.

Finally, asymptotic guarantees can also hide local failure modes. Even algorithms with good classical regret can pay large regret bursts exactly when they begin exploring again after long exploitation phases; in nonstationary or sliding-window views, index policies can then suffer linear sliding regret, while randomized methods may avoid that specific weakness (Boone, 2023). And outside standard bandits, the need for exploration can change sharply: in multi-armed sampling problems, very small exploration on the order of O(\log T) can already give near-optimal regret, whereas no exploration worsens the rate by more than logarithmic factors, showing that the tight link between greedy failure and regret is problem-class specific rather than universal (Pedramfar et al., 2025). So the main takeaway is simple: in ordinary bandits, asymptotic regret guarantees are really guarantees about continued information acquisition. If the algorithm does not keep weakly sampled arms alive long enough, the best possible long-run story usually collapses back to linear regret.

Evidence

(Lee et al., 2025)

Infrequent Exploration in Linear Bandits

> *"greedy approach typically incurs linear regret due to insufficient exploration and inadequate information acquisition (e.g., Example 1 in Jedor et al. [2021])"*
>
> *".While intuitive and computationally efficient, ε-greedy policies are theoretically known to incur suboptimal regret"*
>
> *".Despite their simplicity and intuitive appeal, ETC methods typically result in suboptimal regret compared to fully adaptive exploration strategies such as UCB and TS"*
>
> *".As shown in Table 1, an excessively small number of explorations may result in exponential growth of the constant term with respect to d/Δ, which may significantly degrade the algorithm's finite-time performance."*

(Ameur et al., 2020)

Autonomous Power Decision for the Grant Free Access MUSA Scheme in the mMTC Scenario

"for a constant exploration parameter , the system regret evolves linearly overtime instead of being logarithmic. On the one hand, for a high value, i.e., ≈ 1, the user will continue to only explore random arms even if it came out with the best arm, and on the other hand, for a low value, i.e., << 1, the algorithm will tend to exploit all the time even if it has not sufficiently explored the other arms. In both cases, an important performance loss will be experienced. Therefore, the value is a critical parameter. A revised version called - decreasing greedy has been proposed, where the exploration probability is decreasing toward zero over time with a rate of 1 t . This allows one to essentially explore at the beginning of the learning and mostly to exploit the best arm found so far after a certain amount of time. The new exploration probability is defined as (Adjif et al., 2019)(Auer et al., 2002):"

(Adjif et al., 2019)

Joint Channel Selection and Power Control for NOMA: A Multi-Armed Bandit Approach

"The scarcity of the resources and the steady increasing number of connected devices have spotlighted Non-Orthogonal Multiple Access (NOMA) techniques in order to enable a massive connectivity, where the receiver use Successive Interference Cancellation (SIC) in order to separate multiple users' signal. In this paper, we focus on the uplink transmission in power domain NOMA. We investigate the joint resource allocation and power control problem, and we propose a solution based on Multi-Armed Bandit (MAB). Furthermore, we investigate the exploration-exploitation tradeoff in the proposed reinforcement learning to allow users to choose the appropriate resource block and power level in a distributed manner. The resulting scheme yields better performance compared to the NM-ALOHA and shows an improvement in terms of throughput efficiency."

(Qi et al., 2023)

Forced Exploration in Bandit Problems

"we propose a novel algorithm alternating between greedy rule and forced exploration"

". If the exploration probability satisfies t t − 1 3 , this greedy strategy achieves regret bound on the order of O(T 2 3). Similar to our method, this strategy needs to decide whether to explore at each time step. The difference is that t -greedy is non-adaptive since it does not adapt their exploration schedule to the history."

(Basu et al., 2019)

Blocking Bandits

*"As the UCB Greedy algorithm plays the current best arm, it gets blocked, enforcing the play of the next suboptimal arm-a phenomenon we call free exploration. Free exploration ensures that upto a time horizon t, certain number of arms, namely K * (defined below), are played ct amount of time each, for c > 0, w.h.p"*

". In contrast, we get a O(K 2 Δ(K,1) log T) regret bound when free exploration is ignored."

(Boone, 2023)

The Sliding Regret in Stochastic Bandits: Discriminating Index and Randomized Policies

"We show that index algorithms (UCB, KL-UCB, IMED, etc.) inevitably have linear sliding regret while randomized algorithms such as TS and MED do not suffer from this issue. The analysis of index algorithms underlines that such algorithms tend to locally behave poorly when exploration starts, i.e., when they switch to picking a suboptimal arm after a period of time they only picked optimal ones. This is why we introduce the regret of exploration, that quantifies the local regret at these critical time instants. We show that index algorithms have suboptimal regret of exploration, in opposition to TS and MED."

(Pedramfar et al., 2025)

Multi-Armed Sampling Problem and the End of Exploration

"The following theorem demonstrates that, unlike bandits, there is barely any need for exploration in multiarmed sampling problems. Specifically, we see that with a negligible exploration of O(log(T)), we obtain near-optimal regret bounds that differ from our lower bounds up to logarithmic terms. Moreover, if there is no exploration at all, then the regret bounds would be multiplied by a factor that grows faster than log(T), but slower than T ε , for any ε > 0. Note that with the same amount of exploration, the Explore-Then-Commit algorithms suffer a linear regret in multi-armed bandit problems"

".Moreover, if we have no exploration, i.e., M = 1, then the same bounds hold if we replace Õ with Ô1/2 ."

Finite-Time Performance and Convergence Trade-offs

Finite-time results often make greedy look better than its asymptotic story, because it can exploit a decent arm quickly and avoid the early cost of broad exploration. But this short-run gain is fragile: too little exploration can trap the algorithm, while too much exploration can delay convergence and create large regret before the long-run guarantees help. (6 sources)

Finite-time behavior is where the exploration trade-off becomes most visible. In many experiments, pure

greedy looks strong at first because it exploits immediately, but then it stalls at a positive regret level once it has locked onto an arm that is good enough but not best; methods with continued exploration may start slower yet eventually overtake it because they keep learning (Martin et al., 2020). This is why convergence speed matters as much as asymptotic order in practice: adding some exploration can greatly improve over pure greedy, but naive exploration can still converge slowly. In particular, small-\epsilon and moderate-\epsilon greedy improve on \epsilon=0, yet their convergence can remain poor, whereas more targeted exploration such as Thompson sampling can drive per-period regret toward zero much faster (Abad et al., 2019) (Abad et al., 2018).

The hard part is that the finite-time optimum is not at either extreme. Under-exploration leaves the learner stuck with early errors, but over-exploration also creates large regret by spending too many rounds on weak arms. Empirical comparisons make this very concrete: very small \epsilon can lead to much higher regret because the algorithm does not gather enough corrective samples, while very large \epsilon also performs badly because it keeps exploring after the ranking is already mostly clear; intermediate values can do much better (Wolf, 2025). The same pattern appears in explore-then-commit style tuning: too few exploration rounds produce underexploration, but very many exploration rounds also raise regret, so the best finite-time choice is a balance rather than “more is better” (Wolf, 2025).

A further complication is that finite-time rankings can reverse across problem classes. In many-armed settings, standard UCB can spend too much effort checking too many arms, so broad optimism may have worse short-run regret than simpler rules. Bayati et al. show that even greedy can perform very well in this regime, and that subsampling arms first can make both UCB-style and greedy-style methods much stronger by limiting wasteful exploration; in fact, subsampled greedy can be rate-optimal for Bernoulli many-armed bandits when k>\sqrt{T} (Bayati et al., 2020). Likewise, Jedor et al. report cases where a naive greedy rule beats a learning algorithm for a long time horizon before the latter’s advantage appears, and where a greedier method with enough arms can benefit from “free exploration” and perform surprisingly close to an oracle (Jedor et al., 2020). So finite-time performance is not just about whether an algorithm explores, but also about whether the problem naturally supplies information through many arms, random availability, or other structure.

The practical lesson is that convergence trade-offs are really horizon trade-offs. If the horizon is short or the arm set is very large, aggressive early exploitation can look excellent for a long time; if the horizon is long enough, failure to correct early mistakes usually dominates. Better-performing methods in finite time are therefore often the ones that make exploration more selective rather than simply more frequent: they avoid greedy lock-in without paying the full cost of blind random exploration (Martin et al., 2020) (Abad et al., 2019) (Bayati et al., 2020).

Evidence

(Martin et al., 2020)

Efficient exploration of zero-sum stochastic games

> *"As expected, the greedy exploration method, while initially performing well, gets stuck at positive regret and is eventually outpaced by all the other methods. The random and mincount methods attain a similar level of performance. The epsilon-greedy method (in this case with ε = 0.1)"*

> *"unlike the purely-greedy method, does not get stuck at a positive exploitability. UCB1 performed even better. The best-performing methods were Bayes-UCB and Thompson; this is not that surprising since the analogs of these strategies for the cumulative-regret version of the multi-armed bandit problem are known to perform well in that setting. It is also known that regret bounds for UCB can be converted into Bayesian regret bounds for Thompson sampling [25]."*

(Abad et al., 2019)

Optimal Finite Horizon Sensing for Wirelessly Powered Devices

> *"The greedy algorithm (= 0) has the worst performance as it does not explore at all. By giving non-zero values for , we can see that 0.05-greedy and 0.1-greedy greatly improve upon the greedy algorithm by performing explorations. However, we see a poor*

performance regarding their convergence rate. TS improves the convergence rate significantly by simply adding intelligence to the exploration phase. This makes the TS algorithm to approach a per-period regret of 0 considerably faster than the -greedy algorithm."

(Abad et al., 2018)

Finite Horizon Throughput Maximization and Sensing Optimization in Wireless Powered Devices over Fading Channels

"The greedy algorithm (ε = 0) has the worst performance as it does not explore at all. By giving non-zero values for ε, we can see that 0.05-greedy and 0.1-greedy greatly improve upon the greedy algorithm by performing explorations. However, we see a poor performance regarding their convergence rate. TS improves the convergence rate significantly by simply adding intelligence to the exploration phase. This makes the TS algorithm to approach a per-period regret of 0 considerably faster than the ε-greedy algorithm."

(Wolf, 2025)

A Framework for Fair Evaluation of Variance-Aware Bandit Algorithms

"Smaller (e.g., 10) or much larger (100,000) numbers of exploration rounds lead to high regret due to under-or overexploration"

".Among -Greedy strategies, = 0.05 balances exploration well (regret 301.55, suboptimal rate 0.06031). Greedy with = 0.005 explores too little (regret 1,762.59), while Greedy with = 0.5 explores too much (regret 1,269.05)"

(Bayati et al., 2020)

The Unreasonable Effectiveness of Greedy Algorithms in Multi-Armed Bandit with Many Arms

"The standard UCB algorithm can perform quite poorly in the many-armed regime, because it over-explores arms: even trying every arm once leads to a regret of Ω(k). Instead, we see that the Ω(✓ T) bound is achieved (up to logarithmic factors) by a subsampled upper confidence bound (SS-UCB) algorithm, where we first select m (e.g., m = ✓ T) arms uniformly at random, and then run a standard UCB algorithm (Lai andRobbins 1985, Auer et al. 2002) with just these arms"

".Notably, the greedy algorithm (Greedy)-an algorithm that pulls each arm once and thereafter pulls the empirically best arm for all remaining times-performs exceptionally well. This is despite the well-known fact that the Greedy algorithm can suffer linear regret in the standard Multi-Armed Bandit (MAB) problem, as it might too early fixate on a suboptimal arm"

".We prove that the subsampled greedy algorithm is rate-optimal for Bernoulli bandits when k > \sqrt{T}, and achieves sublinear regret with more general distributions."

(Jedor et al., 2020)

Lifelong Learning in Multi-Armed Bandits

"We see that TS has a "linear" regret indicating that it fails to learn a good parameter γ within that time frame. Whereas the regret of ADAUCB is sublinear and the algorithm is thus learning; however it is outperformed for a relatively long period of time by a naive GREEDY, which is stuck to a suboptimal, yet good arm. The most interesting part is that GREEDY(100) performs extremely well; its regret is remarkably close to the one of ORACLE and is even sublinear. This supports the notion of free exploration for a large enough number of arms of the GREEDY algorithm."

Algorithms and Design Responses

The standard fix for greedy failure is simple: make sure every arm keeps getting sampled, either by random exploration, optimism, or posterior sampling. The main design choice is how targeted that exploration is, because better targeting usually gives better finite-time regret and less risk of lock-in. (LLM Memory)

- **Pure greedy**
- Use only as a baseline or when the problem itself gives strong natural exploration. It is fast and simple, but without forced sampling it can stop checking the best arm after bad early luck and then suffer linear regret with positive probability. (Model-Generated)
- **Fixed-\epsilon greedy**
- This is the easiest repair: with probability \epsilon, sample a random arm; otherwise act greedily. It prevents total lock-in, but if \epsilon stays constant it keeps wasting a fixed share of rounds forever, so long-run regret stays linear. (Model-Generated)
- **Decaying-\epsilon greedy**

- A better use of the same idea is to let exploration shrink over time. This can give sublinear regret if the decay is slow enough to keep all arms sampled, but performance depends heavily on tuning the schedule, and weak tuning can still leave poor finite-time behavior. (Model-Generated)
- Explore-then-commit
- Sample arms in an explicit first phase, then switch to the best empirical arm. This directly addresses insufficient early sampling and is easy to analyze, but it is brittle: too short an exploration phase causes early commitment error, while too long a phase wastes pulls. (Model-Generated)
- UCB-style methods
- These add an uncertainty bonus to each arm, so arms with less data look temporarily better and get sampled. This is a more directed answer to the sampling problem than random \epsilon-exploration, and in standard stochastic bandits it gives strong regret guarantees, often logarithmic or near-logarithmic. (Model-Generated)
- Thompson sampling
- Instead of adding an explicit bonus, sample from a posterior or belief state and act according to that sample. This also keeps uncertain arms alive, often gives strong empirical finite-time performance, and avoids some of the waste of uniform random exploration. (Model-Generated)
- Forced exploration floors
- A practical design rule is to impose a minimum sampling rate for every arm, at least early on. This does not specify one algorithm by itself, but it is the core fix for the failure event behind greedy lock-in: do not let an arm be ruled out before it has enough data. (Model-Generated)
- Adaptive or problem-structured exploration
- If the setting has side information, blocking, arm unavailability, or contexts that naturally reveal other arms, the algorithm can use that structure instead of paying for blind exploration. In such cases, "free exploration" or identifiability from the environment can make greedier methods much safer than in plain bandits. (Model-Generated)
- Many-armed design tricks
- When the number of arms is very large, broad exploration over all arms can be too costly. A common response is to first subsample or screen arms, then run a learning rule on the reduced set; this can improve finite-time performance and can even make greedy-style methods competitive in many-armed regimes. (Model-Generated)
- Nonstationary variants
- If arm values can change, use recency weighting, sliding windows, or explicit re-exploration. The design goal is not just "sample enough" but "sample again soon enough," because stale estimates can create a second form of greedy lock-in even after earlier learning was correct. (Model-Generated)
- Practical rule of thumb
- If you want a safe default, use UCB or Thompson sampling rather than pure greedy. If you still want a greedy-style method for simplicity, use a decaying exploration schedule or another mechanism that guarantees continued sampling, and check finite-time sensitivity to the horizon, gap size, and number of arms. (Model-Generated)